Trader Behavior vs Market Sentiment

A Behavioural Analysis Using Hyperliquid Trading Data

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1 Introduction

Market sentiment plays a critical role in shaping trader behavior, especially in highly volatile cryptocurrency markets. This report analyzes how trader profitability, risk exposure, trading volume, and directional bias vary across different market sentiment regimes using historical trader data from Hyperliquid and the Bitcoin Fear & Greed Index.

The goal is to identify behavioral patterns and hidden signals that may inform more disciplined trading strategies.

2 Datasets

2.1 Bitcoin Fear & Greed Index

The sentiment dataset categorizes each trading day into one of five regimes:

- Extreme Fear
- Fear
- Neutral
- Greed
- Extreme Greed

An ordinal sentiment score was assigned ranging from -2 (Extreme Fear) to $+2$ (Extreme Greed).

2.2 Hyperliquid Trader Data

The trader dataset contains over 200,000 executed trades with information on execution price, trade size, position direction, starting position, and closed profit or loss (PnL).

3 Methodology

3.1 Data Alignment

Trader timestamps were converted to daily granularity and merged with daily sentiment labels based on trade date.

3.2 Feature Engineering

Due to the absence of explicit leverage data, the following behavioral proxies were constructed:

- **Trade Volume (USD):** Absolute trade size in USD
- **Profitability Flag:** Indicator of whether a trade closed in profit
- **Position Exposure:** Proxy for capital at risk using starting position and trade size
- **Directional Bias:** Buy (+1) and Sell (-1) encoding

4 Results

4.1 Average Profitability by Market Sentiment

Market Sentiment	Average Closed PnL
Extreme Fear	34.54
Fear	54.29
Neutral	34.31
Greed	42.74
Extreme Greed	67.89

Table 1: Mean Closed PnL across sentiment regimes

Profitability increases during optimistic regimes, with the highest average PnL observed during Extreme Greed. However, gains are not strictly monotonic across sentiment levels.

4.2 Win Rate Analysis

Market Sentiment	Win Rate
Extreme Fear	0.371
Fear	0.421
Neutral	0.397
Greed	0.385
Extreme Greed	0.465

Table 2: Probability of profitable trades by sentiment

Win rates improve during Extreme Greed, suggesting higher success probability under optimistic market conditions.

4.3 Trading Volume and Risk Exposure

Market Sentiment	Avg Trade Volume (USD)	Avg Exposure Score
Extreme Fear	5349.73	58507.33
Fear	7816.11	84979.19
Neutral	4782.73	92024.53
Greed	5736.88	198512.82
Extreme Greed	3112.25	59701.85

Table 3: Trade volume and exposure by sentiment

Risk exposure peaks during Greed phases, indicating increased capital commitment even when average trade volume does not scale proportionally.

4.4 Directional Bias

Market Sentiment	Directional Bias
Extreme Fear	-0.127
Fear	-0.024
Neutral	0.103
Greed	-0.122
Extreme Greed	-0.165

Table 4: Average directional bias (positive = long, negative = short)

Traders exhibit a mild short bias during extreme sentiment regimes, particularly during Extreme Greed.

4.5 Correlation Analysis

	Sentiment	Volume	Exposure	PnL
Sentiment Score	1.000	-0.032	0.023	0.006
Trade Volume	-0.032	1.000	-0.015	0.124
Exposure Score	0.023	-0.015	1.000	-0.002
Closed PnL	0.006	0.124	-0.002	1.000

Table 5: Correlation matrix of sentiment and trader behavior

The weak direct correlation between sentiment and profitability suggests that sentiment primarily influences risk-taking behavior rather than guaranteeing returns.

5 Hidden Behavioral Trends

Beyond numerical averages, the analysis reveals several intuitive behavioral patterns that help explain how traders react emotionally to market sentiment.

5.1 Higher Profits During Extreme Greed Come With Higher Risk

Traders achieve the highest average profits and win rates during periods of Extreme Greed. However, this increase in profitability is driven by larger risk exposure rather than improved trading accuracy. Even in these optimistic conditions, more than half of trades remain unprofitable, indicating that elevated gains are accompanied by heightened downside risk.

5.2 Fear Increases Activity, Not Caution

Contrary to common belief, trading activity does not decline during Fear regimes. Instead, average trade volume increases, suggesting that traders actively respond to fear through panic exits, aggressive reversals, or attempts to recover losses. This behavior results in higher participation but not necessarily improved outcomes.

5.3 Risk Appetite Peaks During Greed, Not Fear

Position exposure is highest during Greed phases, highlighting that traders are most willing to commit large capital amounts when market confidence is high. This overconfidence-driven behavior makes Greed periods particularly vulnerable to sharp drawdowns if market conditions reverse unexpectedly.

5.4 Market Sentiment Does Not Guarantee Profitability

Correlation analysis shows that market sentiment has almost no direct relationship with trade profitability. While sentiment strongly influences how much traders trade and how much risk they take, it does not meaningfully improve trading success. This reinforces the idea that sentiment affects behavior, not skill.

5.5 Contrarian Bias During Emotional Extremes

Directional analysis indicates a mild short bias during both Extreme Fear and Extreme Greed. This suggests that traders frequently attempt to anticipate market tops and bottoms during emotionally charged periods, often trading against prevailing trends rather than following them.

6 Conclusion

The analysis demonstrates that trader behavior varies significantly across market sentiment regimes. While optimistic sentiment encourages higher participation and risk exposure, profitability does not scale proportionally, highlighting the dangers of sentiment-driven trading. These findings emphasize the importance of disciplined risk management, particularly during euphoric market phases.